Mongolia International Mathematics Competition 2026

Ulaanbaatar City, 22nd to 27th July 2026

Invitational World Youth Mathematics Intercity Competition

Team Contest

Time limit: 70 minutes

Information:

- You are allowed 70 minutes for this paper, consisting of 10 questions printed on separate sheets. For questions 1, 3, 5, 7 and 9, only numerical answers are required. For questions 2, 4, 6, 8 and 10, full solutions are required.
- Each question is worth 40 points. For odd-numbered questions, no partial credits are given. There are no penalties for incorrect answers, but you must not give more than the number of answers being asked for. For questions asking for several answers, full credit will only be given if all correct answers are found. For even-numbered questions, partial credits may be awarded.
- Diagrams shown may not be drawn to scale.

Instructions:

- Write down your team number and team's name in the space provided on every question sheet.
- Enter your answers at the bottom right corners of the corresponding question sheets.
- During the first 10 minutes, the four team members examine the first 8 questions together and discuss them. After 10 minutes they distribute the questions among themselves, with every team member allotted at least 1 question.
- During the next 35 minutes, the four team members write down the solutions of their allotted problems on the respective question sheets, with no further communication or discussion.
- During the last 25 minutes, the four team members work together to write down the solutions of the last 2 questions on the respective question sheets.
- It is forbidden to use instruments such as protractors, calculators and electronic devices, smart watches.
- At the end of the contest, you must hand in the envelope containing all question sheets and all scrap paper.

English Version

Team: _____ Team No.: _____

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1. In preparation for the annual Naadam Festival in the Mongolian highlands, the local organizing committee is arranging the distribution of ***Buuz*** (traditional Mongolian steamed dumplings) for the honored guests representing a specific district.



The committee uses two types of food containers: small boxes and large boxes. It is known that:

- The capacity of each type of box is a positive integer number of dumplings.
- The capacity of one small box is less than half the capacity of one large box.
- One large box and one small box together hold exactly 63 dumplings.
- One large box and two small boxes together hold only two-fifths of the total number of dumplings prepared for all guests.

If the committee tries to pack all the dumplings prepared into 3 large boxes and 2 small boxes, they find that n dumplings remains unpacked, where n is a positive integer. What is the sum of all possible values of n?

Answer: _____

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2. It is known that there exists a positive integer x such that each of the following numbers:

x, $2x$, $3x$, $4x$ and $5x$ is a perfect power.

Provide one example of such an integer x and explain how you obtained it.

(Note: a^i is a perfect power where a and i are positive integers and $i > 1$.)

One example provided is as follows:

Answer: $x =$, $2x =$, $3x =$

$4x =$, $5x =$

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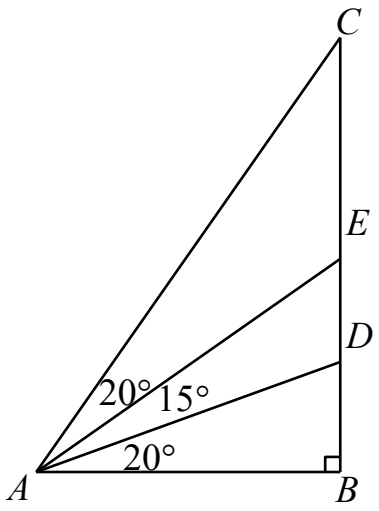
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3. As shown in the diagram below, ABC is a right triangle. Points D and E lie on side BC such that $\angle BAD = 20°$, $\angle DAE = 15°$ and $\angle EAC = 20°$.
If $EC = 4$ cm, what is the length, in cm, of BD?



Answer: _____ cm

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4. Let x, y and z be real numbers, not all equal, such that

$$x+y+z=0 \text{ and } x^3+y^3+z=y^3+z^3+x=z^3+x^3+y.$$

Prove that $\dfrac{(x^2+y^2+z^2)^2}{x^4+y^4+z^4}=2$.



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5. The numbers 1, 2, 3, …, 32 and 33 are written on a board. A student repeatedly performs the following operation: erase two numbers such that one is a multiple of the other, and write down their integer quotient on the board. The process stops when no number on the board is a multiple of another. At least how many numbers remain on the board?

Answer: _____

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6. Let a, b, c and d be positive real numbers such that

$$(a+4c)(b+d)=2(ac+2bd).$$

What is the minimum possible value of the sum $S=\frac{a}{b}+2\frac{b}{c}+4\frac{c}{d}+8\frac{d}{a}$?

Answer: _____

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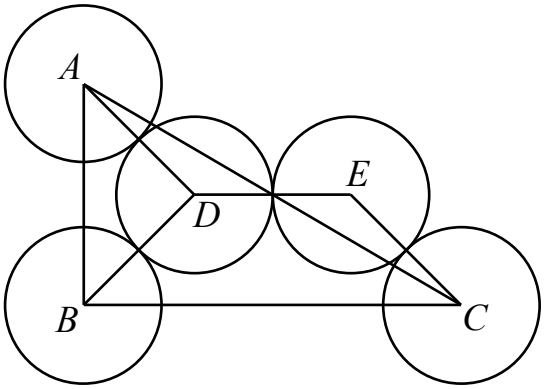
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7. As shown in the diagram below, A, B, C, D and E are the centres of five circles, each with radius 1 cm such that $\angle ABC = 90°$, $\angle BAC = 60°$ and $BC // DE$. Furthermore, the pairs of circles centered at (A, D), (B, D), (D, E) and (E, C) are externally tangent to each other and no other pair of circles have any intersections. Suppose the maximum possible distance between two points chosen on any of these five circles can be expressed as $a + \sqrt{b} + \sqrt{c}$ cm, where a, b and c are positive integers and $b < c$. What is the value of $a + 10b + 100c$?



Answer: _____

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8. A sidewalk of integer length N units, where $1 \le N \le 1000$, is to be paved using tiles whose lengths are the Fibonacci numbers: 1, 1, 2, 3, 5, 8, …. The tiles are placed in increasing order of length, starting from the first 1-unit tile. Whenever the next Fibonacci tile is longer than the remaining uncovered length of the sidewalk, the sequence restarts from the next available 1-unit tile.

Each start or restart of the sequence from the value of 1 defines a ***new sequence*** (thus the first tile on the sidewalk marks the beginning of the first sequence). If N is the smallest positive integer that requires the most sequences, how many tiles of any size, in total, are used?

For example, the diagram below illustrates the paving of a sidewalk of length $N = 26$ with 11 tiles, for which 3 sequences were required (marked by the 3 gray tiles).

(**Note:** The initial two numbers of Fibonacci sequence are both 1 and then each number in the sequence is the sum of the two numbers preceding it.)



Answer: _____ tiles

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9. A subset S of {1, 2, 3, …, 2025, 2026} is called *good*, if for any three elements of S, their least common multiple also belongs to S. For example, {3, 9, 12, 36} is *good*. What is the maximum possible number of elements in a *good* subset of {1, 2, 3, …, 2025, 2026}?

Answer: _____

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10. As shown in the diagram below, triangle ABC is an acute triangle with altitudes AX, BY and CZ. Let P be the intersection point of XY and CZ. The line through P parallel to AX intersects side AC at T and line XZ at Q. Let H be the orthocentre of triangle ABC. Prove that $\angle HCB = \angle BCQ$.

